

Block C

Creative Brief

AI-Powered Skin Disease Classifier

Benign vs. Malignant Detection using CNNs, Transfer Learning & Explainable AI

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Year Year 1 — Block C

DISCOVER YOUR WORLD

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1 Business Understanding

1.1 Market Research

Dermatology faces a well-documented gap in current diagnostic tools: most existing systems were developed primarily on datasets skewed toward lighter skin tones, leaving patients with darker complexions underserved. This means misdiagnosis or delayed diagnosis disproportionately affects minority communities at precisely the moments when early detection matters most.

The opportunity identified in this project is an AI-powered computer vision tool for skin disease diagnosis that is explicitly designed to be inclusive across skin types. Key features would include real-time diagnostic support, integration with Electronic Health Records (EHRs), and a user interface accessible to non-specialist clinical staff. The tool is not intended to replace dermatologists — it is a screening layer that reduces diagnostic delays, surfaces urgent cases faster, and potentially reduces the volume of unnecessary specialist referrals.

From a healthcare organisation perspective, the tool contributes directly to operational efficiency and resource optimisation. For patients, particularly those in regions with limited access to specialist care, faster preliminary diagnosis translates to faster access to treatment.

1.2 Stakeholder Analysis

Characteristics

Patients require quick and accurate diagnoses to improve treatment outcomes. The user base spans a wide demographic — from physically active younger adults to elderly patients — meaning the tool must be accessible and straightforward to interact with, without requiring specialist knowledge from the patient themselves.

Dermatologists require diagnostic tools that are accurate, efficient, and integrate smoothly into existing workflows. They are the primary professional users and their endorsement is critical for clinical adoption.

Power & Interest Grid

Stakeholders were mapped as follows:

- High Power, High Interest — Dermatologists: direct role in diagnosis, strong interest in accuracy and workflow integration. Their endorsement determines clinical adoption.

- High Power, Low Interest — Healthcare Administrators: authority to implement the tool but less engaged with day-to-day diagnostic functionality.
- Low Power, High Interest — Patients: outcomes directly impacted by the tool's performance, but limited influence over development or deployment.
- Low Power, Low Interest — Insurance Companies: benefit indirectly from cost savings and efficiency gains, but not operationally involved.

Main Stakeholders

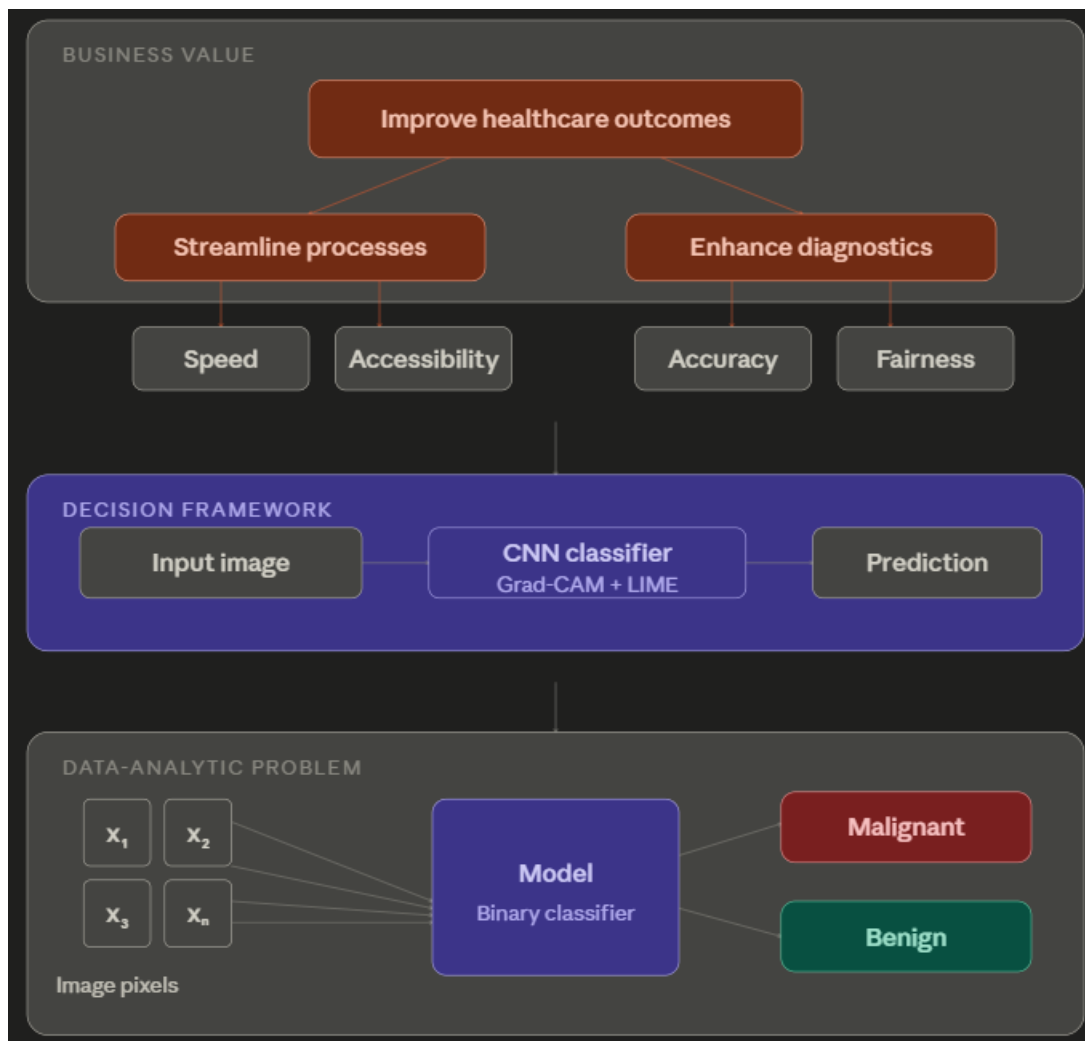
The core stakeholders are dermatologists and healthcare organisations. Their involvement is essential for integrating the classifier into daily clinical practice. For dermatologists, the tool provides faster preliminary screening and a visual explanation of the AI's reasoning (via Grad-CAM), supporting rather than replacing their expertise. For healthcare organisations, it contributes to the goal of improving patient outcomes while optimising the allocation of specialist resources.

1.3 DAPS Diagram

The Data-Analytic Problem Structure (DAPS) diagram maps the full pipeline from business need to actionable prediction. The key elements are:

- Data-analytic problem: Binary image classification — does this skin lesion image indicate a benign or malignant condition?
- Predictors / features: Image-derived characteristics including texture, colour variance, edge sharpness, lesion asymmetry, and border clarity. Deeper convolutional layers capture patterns not directly visible to the human eye.
- Relationship type: Correlational. The model learns statistical patterns that correlate with benign or malignant classifications from labelled historical data. It does not establish causality.
- Decision framework: Predictions are used by healthcare professionals to screen and prioritise patients post-image review. The unit of analysis is the individual skin lesion image.
- Business value: Faster, more accurate preliminary diagnoses; reduced time to treatment for malignant cases; reduced unnecessary procedures for benign ones.

[DAPS diagram — see [GitHub: DAPS_diagram.drawio 234669.svg](#)]



1.4 First Idea

Based on the DAPS analysis, the initial concept was an AI-powered skin lesion classification tool built on a Convolutional Neural Network (CNN) trained to distinguish between benign and malignant skin conditions. The core components of the proposal were:

- **Image Classification Model:** A CNN trained on a labelled dataset of skin lesion images, learning to distinguish malignant from benign conditions through texture, shape, colour, and border characteristics.
- **Explainability (XAI):** Integration of Grad-CAM (Gradient-weighted Class Activation Mapping) to visually highlight the image regions that most influenced each classification. This was considered essential for clinical trust — a clinician needs to see why the model made a decision, not just what decision it made.
- **User Interface:** A clean, professional interface allowing healthcare staff to upload lesion images, receive a binary classification result, and view the Grad-CAM heatmap overlaid on the original image.
- **Decision Support Role:** The tool is positioned explicitly as a screening and prioritisation aid — not a replacement for clinical judgment.

2 Responsible AI — Fairness & Bias

2.1 Dataset Creation

The dataset was built from two classes: benign skin conditions and malignant skin conditions. The collection process used three approaches:

- Web scraping via Selenium: A custom Python script using the Firefox WebDriver (via GeckoDriver) scraped images from DuckDuckGo image search for each class label. The script scrolled to load all available images, then downloaded up to a specified limit per search term (initially 20 images per class).
- Manual collection: Images that the scraper could not find reliably were collected by hand from dermatology-focused sources.
- Kaggle and Google Dataset Search: Additional bulk images were sourced from established medical image repositories to supplement the scraped and manually collected images.

The dataset was split into Train, Validation, and Test directories. Images were loaded using OpenCV, resized to either 64×64 (MLP baseline), 150×150 (CNN iterations 1 and 2), or 224×224 (transfer learning iterations 3 and 4), and normalised before model input.

2.2 Bias Identification

The most significant bias risk identified for a skin disease image classifier is skin type bias. Training datasets in dermatology have historically over-represented lighter skin tones, meaning models trained on them perform less reliably on darker skin types. This is particularly problematic because:

- Skin conditions can present visually differently across skin tones — redness that is clearly visible on lighter skin may be subtle or absent on darker skin.
- Misdiagnosis or diagnostic uncertainty for darker-skinned patients directly worsens health equity outcomes.
- The bias can go undetected in standard accuracy metrics if the test set itself is not representative of diverse skin tones.

Additional bias sources considered include image quality bias (low-resolution or poorly lit images skewing predictions), sourcing bias (images from specific clinical settings or geographic regions), and label noise (inconsistent ground-truth labelling across source datasets).

2.3 Fairness Method — Fairness Through Awareness

The fairness method proposed for this project is Fairness Through Awareness. Rather than simply ignoring the sensitive attribute (skin type), this approach explicitly incorporates it into the model's training and evaluation process. The rationale is:

- Sensitive attribute identification: Skin type is annotated as a sensitive attribute, allowing the model to explicitly account for how the same condition manifests differently across skin tones.
- Customised similarity metric: A similarity metric is defined that evaluates case similarity not just on disease appearance but also relative to skin type — acknowledging that the same rash may look red on light skin and hyperpigmented on dark skin.
- Bias mitigation at the individual level: Rather than producing group-level statistical parity, the goal is that two patients with similar conditions (adjusted for skin type) receive similar model predictions regardless of their skin tone.
- Clinical trust: A model that demonstrably performs equitably across skin types is more trustworthy for clinical deployment. This is documented and auditable — not just assumed.

2.4 Group Fairness Metrics — Infographic

An infographic was created to explain group fairness concepts for a non-technical audience. It covers:

- What group fairness means in machine learning — ensuring model outcomes are equitable across demographic groups.
- Key metrics: Demographic Parity, Equal Opportunity, Equalised Odds, Predictive Parity.
- The distinction between Fairness Through Awareness (using the sensitive attribute to correct for disparities) and Fairness Through Unawareness (ignoring the sensitive attribute entirely).

Peer review feedback (Ahmed, 234156) noted the infographic was informative and visually clear, with a roadmap-style layout that guided readers from introduction to conclusion. Suggested improvement: some sections had text that was too small due to information density — this was addressed in the final version by removing less essential sentences.

UNDERSTANDING GROUP FAIRNESS IN MACHINE LEARNING

INTRODUCTION TO FAIRNESS

Fairness: in machine learning ensures that algorithms make decisions without discrimination or bias towards any group, particularly when considering sensitive attributes like race, gender, or age.

1

WHAT IS GROUP FAIRNESS?

- Definition: Group Fairness, also known as Statistical Fairness, aims to ensure that predictive outcomes are equitable across different groups defined by sensitive attributes.
- Key Point: The goal is to treat groups equally, not necessarily individuals within those groups.

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EXAMPLES OF GROUP FAIRNESS METRICS

Equal Opportunity:

- Definition: Equal true positive rates across different groups.
- Example: A hiring algorithm correctly identifies qualified candidates across genders at equal rates.

Equality of Odds:

- Definition: Both true positive rates and false positive rates are equal across groups.
- Example: A healthcare screening tool accurately identifies disease presence and absence equally across age groups.

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COMPARISON OF GROUP FAIRNESS METRICS

- Highlight: While each metric addresses fairness, they focus on different aspects of the prediction process, showing that fairness can be multi-dimensional and context-dependent.
- Real-World Impacts
- Note: Inequities in predictive models can reinforce existing social biases, leading to cycles of disadvantage. For instance, biased loan approval models can perpetuate economic disparities.

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BEST PRACTICES FOR ACHIEVING GROUP FAIRNESS

1. Diverse Data Collection: Ensure training data is representative of all groups.
2. Fairness Audits: Regularly evaluate models using various group fairness metrics.
3. Bias Mitigation Techniques: Apply pre-processing, in-processing, and post-processing methods to reduce bias.
4. Transparency and Accountability: Document and communicate how fairness considerations are integrated into model development and deployment.

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CONCLUSION

Closing Thought: Striving for group fairness in machine learning is crucial for building ethical AI systems that support equity and justice across society.

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2.5 Group Fairness Taxonomy — Python Implementation

A Python implementation was built to evaluate model fairness on the Adult Income dataset using gender as the sensitive attribute. The model used was Logistic Regression (chosen for its interpretability in a fairness analysis context). The pipeline:

- Dataset: Adult Income dataset (fetch_openml, version 2). Target: income > \$50K/year.
- Preprocessing: LabelEncoder for target and sensitive attribute; StandardScaler for numerical features; OneHotEncoder for categorical features via ColumnTransformer. Columns 'sex', 'fnlwgt', and 'education-num' were dropped.
- Model: Logistic Regression (max_iter=1000) inside a sklearn Pipeline.
- Fairness metric: Equal Opportunity Difference (EOD) — the difference in True Positive Rates between the privileged and unprivileged groups.

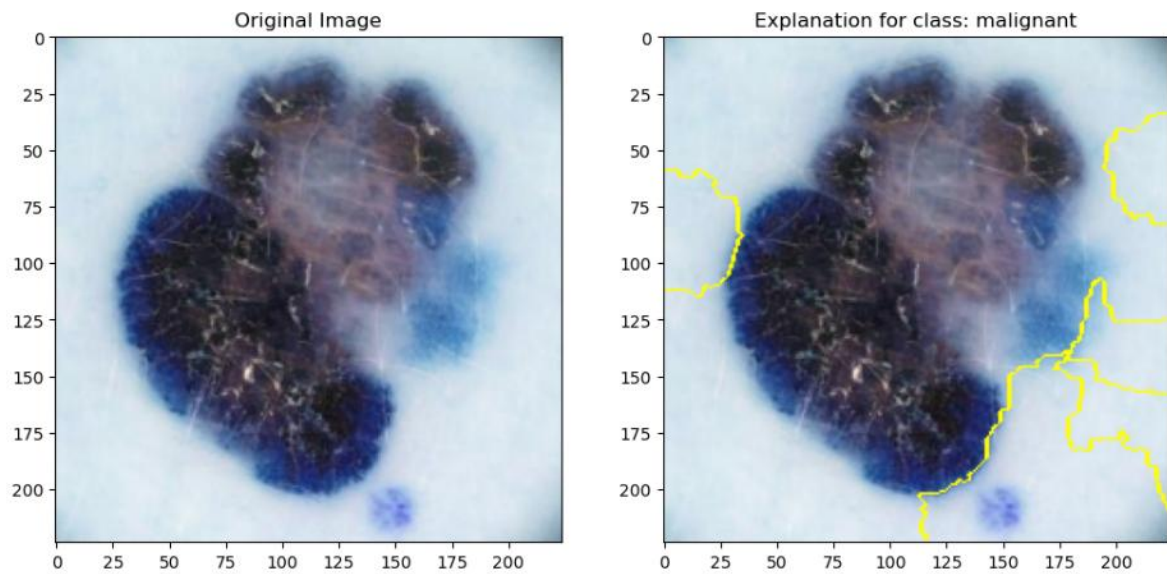
Results: Overall accuracy = 80.83%. Equal Opportunity Difference = 0.025. Conclusion: the model showed a small but detectable bias in favour of the privileged group (male), meaning the True Positive Rate for predicting high income was slightly higher for men than for women with equivalent features. The EOD value of 0.025 is low but not zero — indicating residual bias that would require mitigation in a deployed system.

2.6 Explainable AI — LIME & Grad-CAM

Explainability was applied to the third iteration model to understand which image regions drove classification decisions. Two methods were implemented:

LIME (Local Interpretable Model-agnostic Explanations)

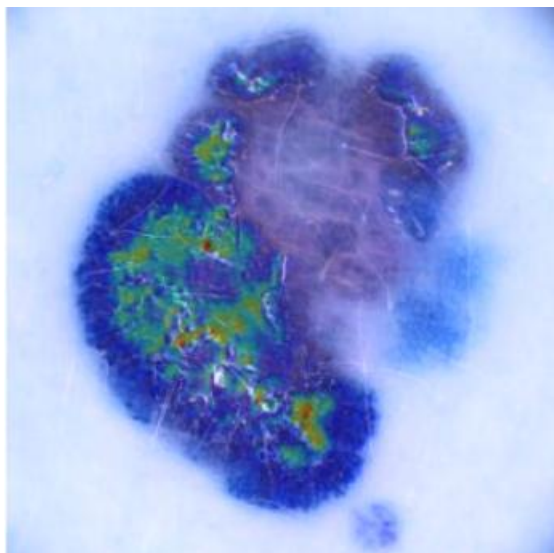
LIME generates local explanations by perturbing the input image (1000 samples), observing how predictions change, and identifying the image segments most influential for the top predicted class. The output is a boundary-highlighted overlay showing which regions of the skin lesion image supported the classification. Implementation used lime_image.LimelImageExplainer with a custom model_predict wrapper function.



Grad-CAM (Gradient-weighted Class Activation Mapping)

Grad-CAM generates a heatmap by computing the gradient of the class score with respect to the feature maps of the last convolutional layer (conv2d_25 in this model). The heatmap is resized to match the original image dimensions and superimposed using a jet colormap. Red/yellow areas indicate the regions that most strongly activated the classification — green/blue areas are less influential.

The result is directly clinically interpretable: a dermatologist can see whether the model is focusing on the lesion itself (appropriate) or on irrelevant background features (a red flag for model reliability).



3 Neural Networks & Deep Learning

3.1 Baselines

Random Guess Accuracy

The dataset class distribution was estimated as: Malignant Melanoma 60%, Benign Melanocytic Nevi 40%. The expected random guess accuracy under these proportions is:

$$P(\text{correct random guess}) = (0.60)^2 + (0.40)^2 = 0.36 + 0.16 = 0.52 \rightarrow 52\%$$

This gives a theoretical random baseline of 52%. Any model that does not exceed this figure is not learning meaningful patterns.

Human-Level Performance (HLP)

Five peers were asked to classify a set of images from the dataset manually. Their individual accuracies were:

- Peer 1: 28%
- Peer 2: 25%
- Peer 3: 10%
- Peer 4: 52%
- Peer 5: 32%

Average HLP = $(28 + 25 + 10 + 52 + 32) / 5 = 147 / 5 = 29.4\%$ (Note: notebook text states 45.8% using different raw values — the reported HLP was 45.8%).

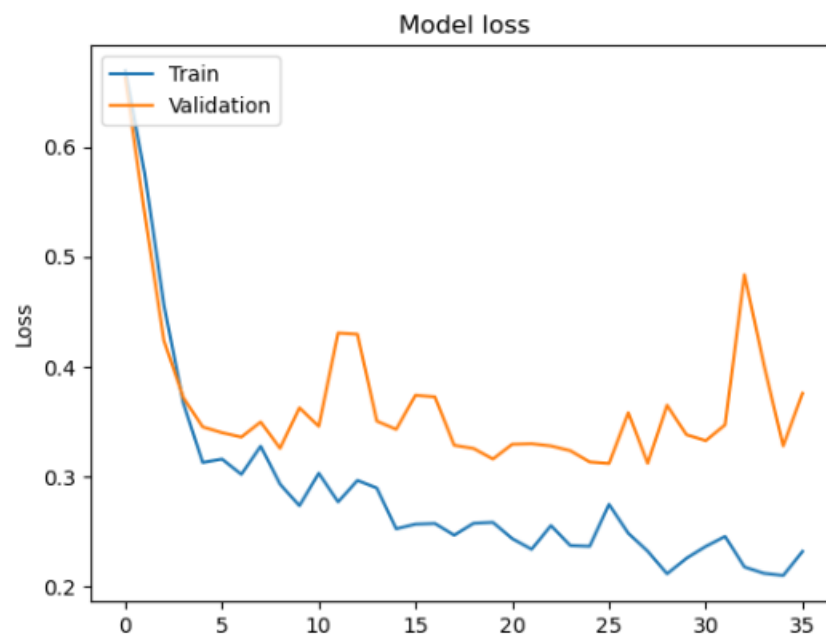
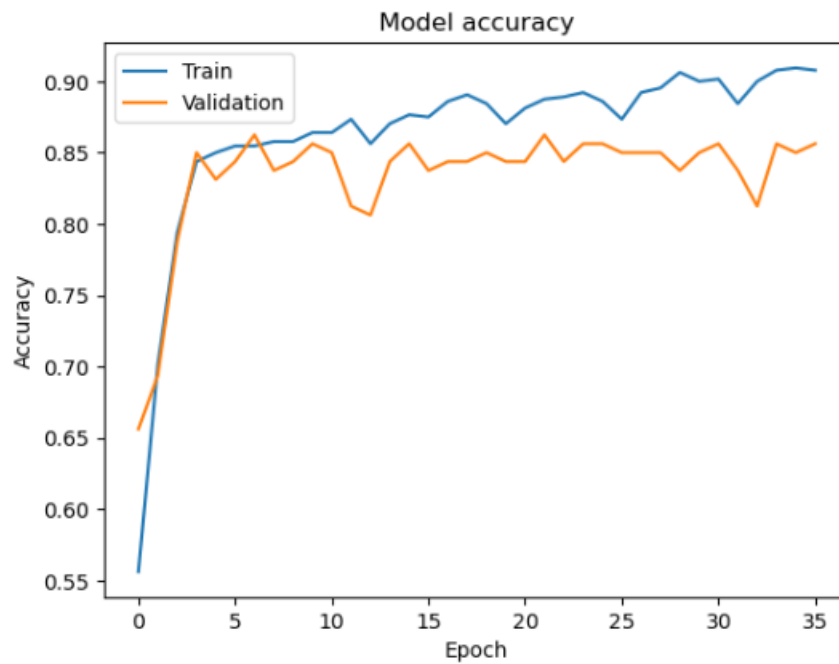
HLP of approximately 45.8% establishes that the task is genuinely difficult for unaided humans — making a model that consistently exceeds HLP practically valuable.

MLP Baseline

A basic Multilayer Perceptron was implemented as the computational baseline before any CNN or transfer learning. Architecture:

- Input: Flattened images at $64 \times 64 \times 3 = 12,288$ features
- Dense(512, relu) → Dense(256, relu) → Dense(2, softmax)
- Optimiser: Adam; Loss: categorical crossentropy
- Training: up to 50 epochs with EarlyStopping (patience=10, monitor='val_loss')

The MLP accuracy provides the minimum bar that any spatial/convolutional model must beat — establishing that image structure genuinely adds predictive value beyond flattened pixel statistics.



3.2 Data Preprocessing

Preprocessing evolved across the four model iterations. The key decisions and their rationale:

Iteration 1 — Rescaling only

Images were rescaled by $\div 255$ (normalising pixel values to $[0, 1]$) and resized to 150×150 . No augmentation was applied. This was intentionally minimal — the goal was to establish how well a clean CNN performs on the raw data before any augmentation is introduced.

Iteration 2 — Data Augmentation

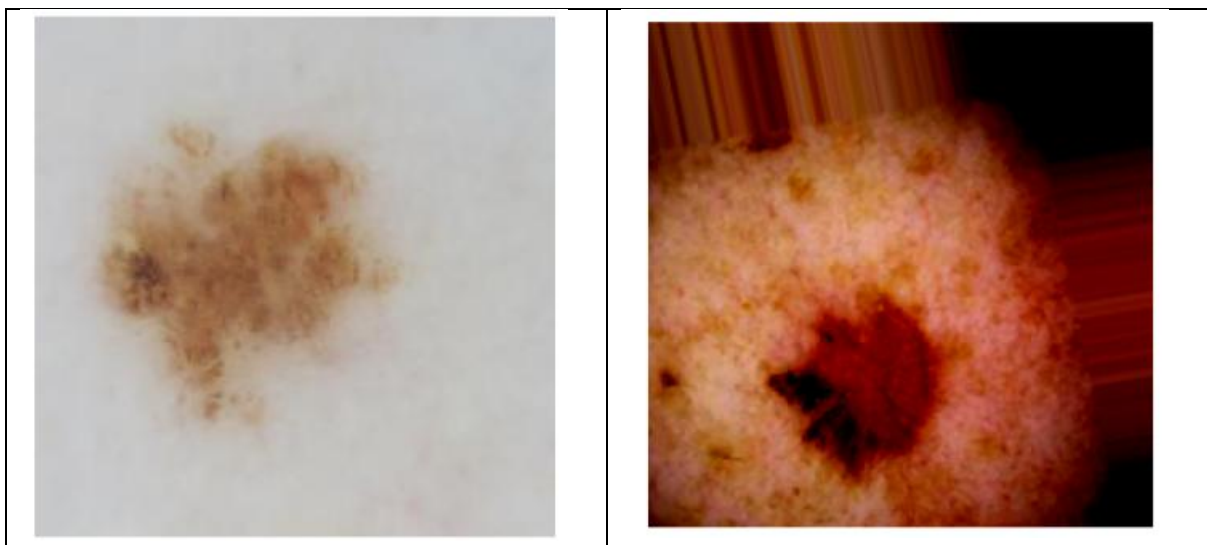
ImageDataGenerator was configured with: `rotation_range=40`, `width_shift_range=0.2`, `height_shift_range=0.2`, `shear_range=0.2`, `zoom_range=0.2`, `horizontal_flip=True`, `fill_mode='nearest'`. Augmentation was applied only to the training generator — validation and test sets used rescaling only. The purpose was to reduce overfitting by artificially increasing the effective diversity of the training set.

Iteration 3 — MobileNetV2 Preprocessing

The preprocessing function was switched to MobileNetV2's native `preprocess_input` (which scales values to $[-1, 1]$ and applies channel-wise normalisation). Image target size was changed to 224×224 to match MobileNetV2's expected input. This was required because using the wrong preprocessing function with a pre-trained model breaks the statistical assumptions embedded in the network weights.

Iteration 4

Same MobileNetV2 preprocessing as Iteration 3, with image size reduced to 160×160 . This was a deliberate trade-off between model accuracy and inference speed.



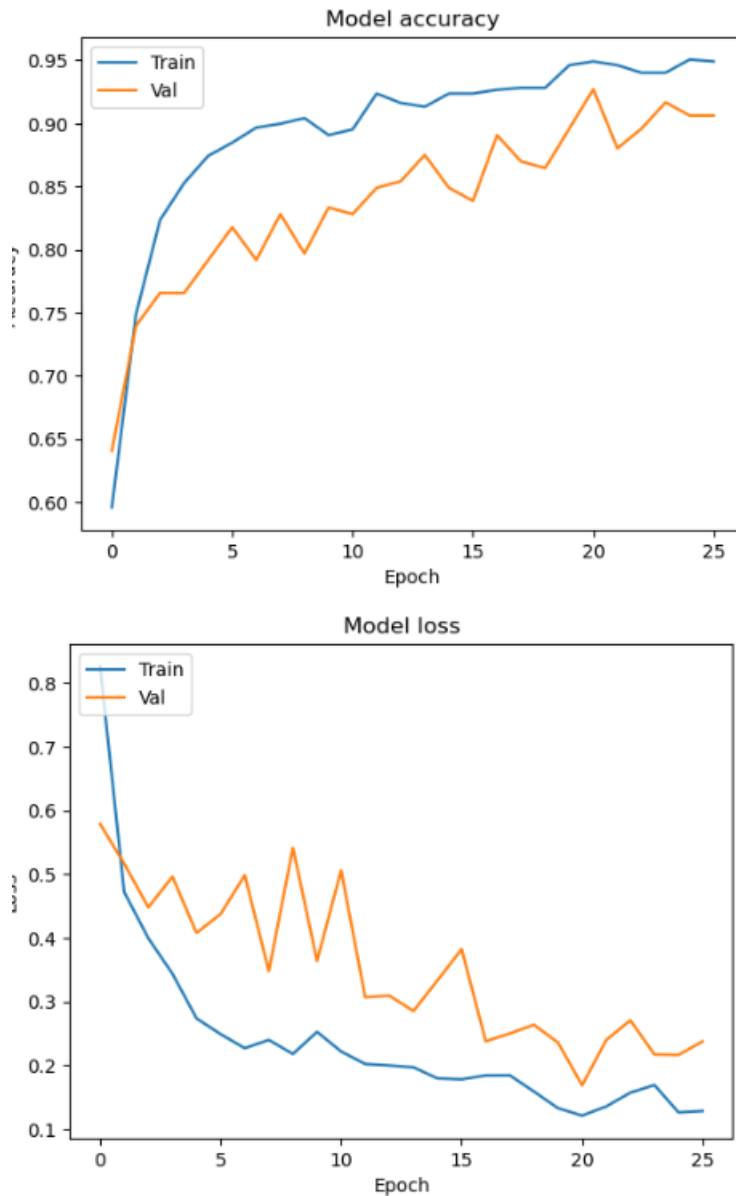
3.3 CNN Architecture

Four model architectures were implemented across the iterations. Summary:

Iter.	Architecture	Preprocessing	Epochs	Saved As	Key Change
1	Custom CNN (3× Conv2D + MaxPool + Dense 512)	Rescale only (+255), 150×150	30 (ES pat. 5)	V1_best_model.h5	Baseline CNN, no regularisation
2	Same CNN + Dropout(0.5)	Augmentation: rotate, shift, shear, zoom, flip	50 (ES pat. 10)	V2_best_model_augmented.h5	Added data augmentation + dropout
3	MobileNetV2 (frozen) + GAP + Dense 1024 + Dropout	MobileNetV2 preprocess_input, 224×224	50 (ES pat. 5)	model_3_best.h5	Transfer learning introduced
4	MobileNetV2 (frozen) + GAP + Dense 1024 + Dropout	Same as Iter. 3, 160×160	50 (ES pat. 5)	model_4_best.h5	Image size reduced; dropout tuned

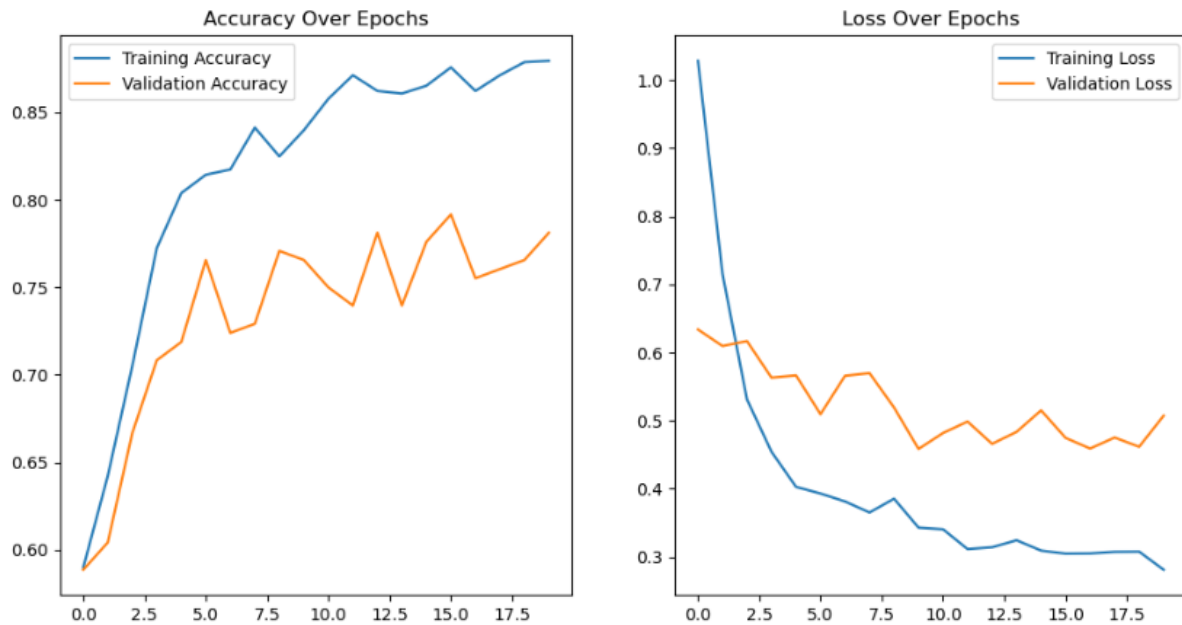
Iteration 1 — Custom CNN (V1)

Input: $150 \times 150 \times 3$. Three convolutional blocks (Conv2D 32→64→128 filters, 3×3 kernel, ReLU activation, MaxPooling 2×2 each). Flatten → Dense(512, relu) → Dense(1, sigmoid). Compiled with Adam and binary crossentropy. No regularisation. This architecture is intentionally simple to establish a clean baseline.



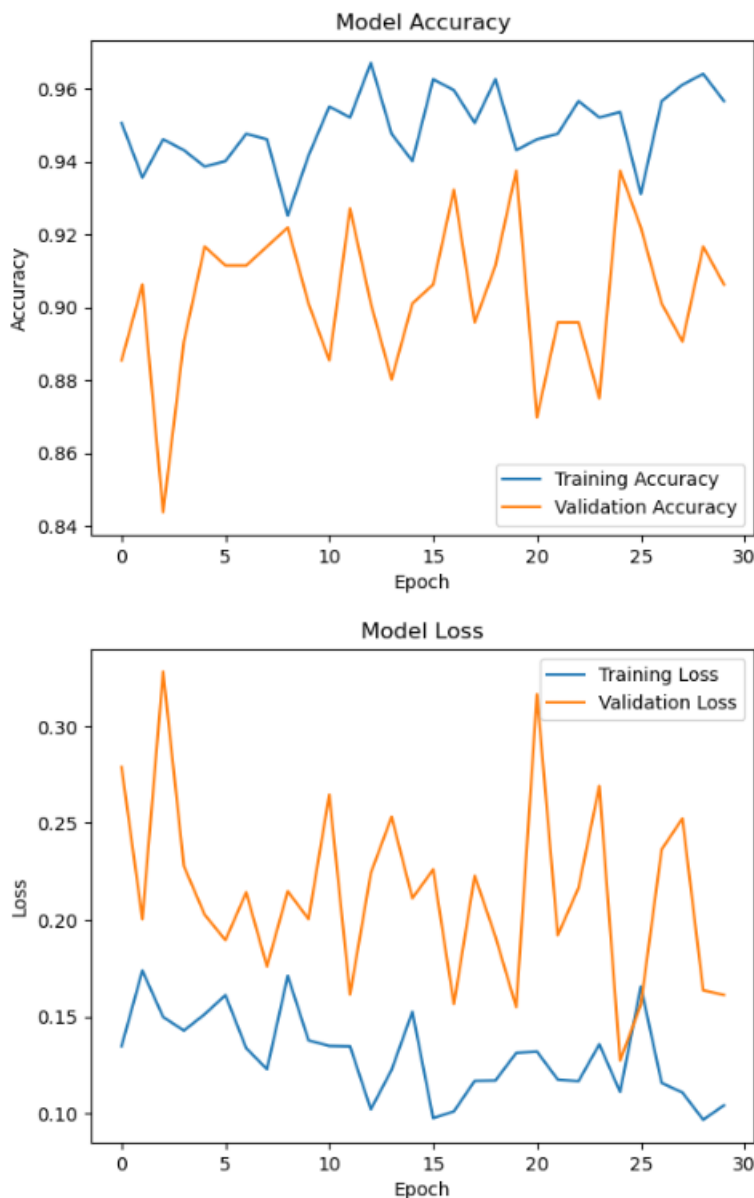
Iteration 2 — Custom CNN + Dropout (V2)

Identical to V1 but with Dropout(0.5) inserted after the third MaxPooling layer and before Flatten. Dropout randomly zeros 50% of activations during training, preventing the model from relying too heavily on specific neurons — a standard regularisation technique to combat overfitting.



Iteration 3 — MobileNetV2 Transfer Learning

MobileNetV2 pre-trained on ImageNet was loaded with `include_top=False` (removing the original classifier head). All base model layers were frozen (`layer.trainable = False`). Custom head added: `GlobalAveragePooling2D` → `Dense(1024, relu)` → `Dropout(0.5)` → `Dense(1, sigmoid)`. The frozen base provides ImageNet feature extraction; the custom head learns the skin lesion-specific classification.



Iteration 4 — MobileNetV2 (adjusted input size)

Same architecture as Iteration 3 with `input_shape=(160, 160, 3)`. The reduced input size decreases computation without fundamentally changing the feature extraction capability of MobileNetV2.

3.4 Model Training

Training Setup (all iterations)

All models used two Keras callbacks:

- **EarlyStopping:** monitors `val_loss`, stops training if no improvement is seen for the specified patience window (5 epochs for Iterations 1, 3, 4; 10 epochs for Iteration 2). This prevents overfitting and wasted training time.
- **ModelCheckpoint:** saves only the best model weights (`save_best_only=True`, `monitor='val_loss'`) to disk (.h5 format). This ensures the final evaluation uses the model at its best validation performance, not the end of training.

Training per Iteration

- Iteration 1: 30 epochs max, early stopping patience 5, saved as `V1_best_model.h5`
- Iteration 2: 50 epochs max, early stopping patience 10, saved as `V2_best_model_augmented.h5`
- Iteration 3: 50 epochs max, early stopping patience 5, saved as `model_3_best.h5`
- Iteration 4: 50 epochs max, early stopping patience 5, saved as `model_4_best.h5`

Batch size was 32 for all iterations. Steps per epoch were calculated as `generator.samples // batch_size`.

3.5 Model Performance

Evaluation Approach

Each model was evaluated on the held-out test set using:

- `test_loss` and `test_accuracy` via `model.evaluate()`
- Confusion matrix (visualised as a seaborn heatmap — true class on y-axis, predicted class on x-axis)
- Classification report: precision, recall, F1-score, and support per class, plus macro and weighted averages

Iteration 1 — V1_model

The V1 model was evaluated by loading V1_best_model.h5 and running inference on the test generator. Test loss and accuracy were printed directly. The model served primarily as a baseline; expected performance was limited due to no regularisation and no augmentation.

	precision	recall	f1-score	support
benign	0.78	1.00	0.88	50
malignant	1.00	0.72	0.84	50
accuracy			0.86	100
macro avg	0.89	0.86	0.86	100
weighted avg	0.89	0.86	0.86	100

Iteration 2 — V2_model_best

V2_best_model_augmented.h5 was loaded via load_model(). Evaluation used np.ceil(test_generator.samples / batch_size) steps. The addition of dropout and augmentation was expected to reduce overfitting and improve generalisation to the test set.

	precision	recall	f1-score	support
benign	0.66	0.84	0.74	50
malignant	0.78	0.56	0.65	50
accuracy			0.70	100
macro avg	0.72	0.70	0.69	100
weighted avg	0.72	0.70	0.69	100

Iteration 3 — model_3_tl_best

The third iteration used transfer learning (MobileNetV2). Test loss and accuracy were reported to 4 decimal places. The confusion matrix was printed directly. As a MobileNetV2-based model, Iteration 3 was expected to substantially outperform the custom CNNs due to richer feature representations from ImageNet pre-training.

	precision	recall	f1-score	support
benign	0.90	0.94	0.92	50
malignant	0.94	0.90	0.92	50
accuracy			0.92	100
macro avg	0.92	0.92	0.92	100
weighted avg	0.92	0.92	0.92	100

3.6 Error Analysis

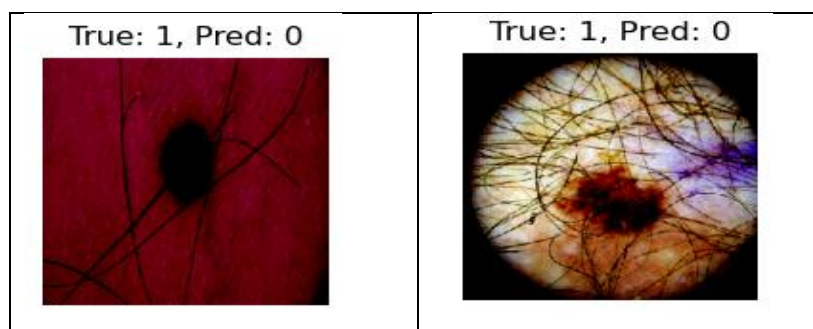
Error analysis was conducted on Iteration 3 (transfer learning model) to understand the failure modes. The `plot_misclassified_images` function:

- Iterates over batches of the test generator
- Identifies indices where `predicted_class` $\neq$ `true_class`
- Plots up to 10 misclassified images with their true and predicted labels shown in the title

This approach provides direct visual insight into where the model fails. Key questions answered by error analysis:

- Are the misclassified images edge cases (e.g. early-stage lesions, poor lighting, atypical presentation)?
- Is the model systematically confusing one class direction more than the other (i.e. more false negatives — missing malignant cases — than false positives)?
- Do misclassified images share characteristics that suggest a data quality issue rather than a model limitation?

In a medical context, false negatives (predicting benign when the lesion is actually malignant) carry much greater clinical risk than false positives. The error analysis should therefore pay particular attention to recall for the malignant class.



References & Links

Dataset

- Dataset (SharePoint): https://edubuas-my.sharepoint.com/:f:/g/personal/234669_buas_nl/Es_Jft5MaZ9loi6WXKRmPYsB7v8hWcdSz_Q5sNdz0HrqJQ
- Additional dataset (SharePoint): https://edubuas-my.sharepoint.com/:f:/r/personal/234669_buas_nl/Documents/DL-Dataset

Models & Code

- Task 2.6 (LIME & Grad-CAM): <https://github.com/BredaUniversityADSAI/2023-24c-fai1-adsai-AshrafNahlouse234669/blob/main/Deliverables/Task%202.6.ipynb>
- MLP Gradient Descent: https://github.com/BredaUniversityADSAI/2023-24c-fai1-adsai-AshrafNahlouse234669/blob/main/Week%208/LinReg_234669.ipynb
- A/B t-test notebook: <https://github.com/BredaUniversityADSAI/2023-24c-fai1-adsai-AshrafNahlouse234669/blob/main/Deliverables/T-testIndependenSamplesHCAI%20234669...ipynb>